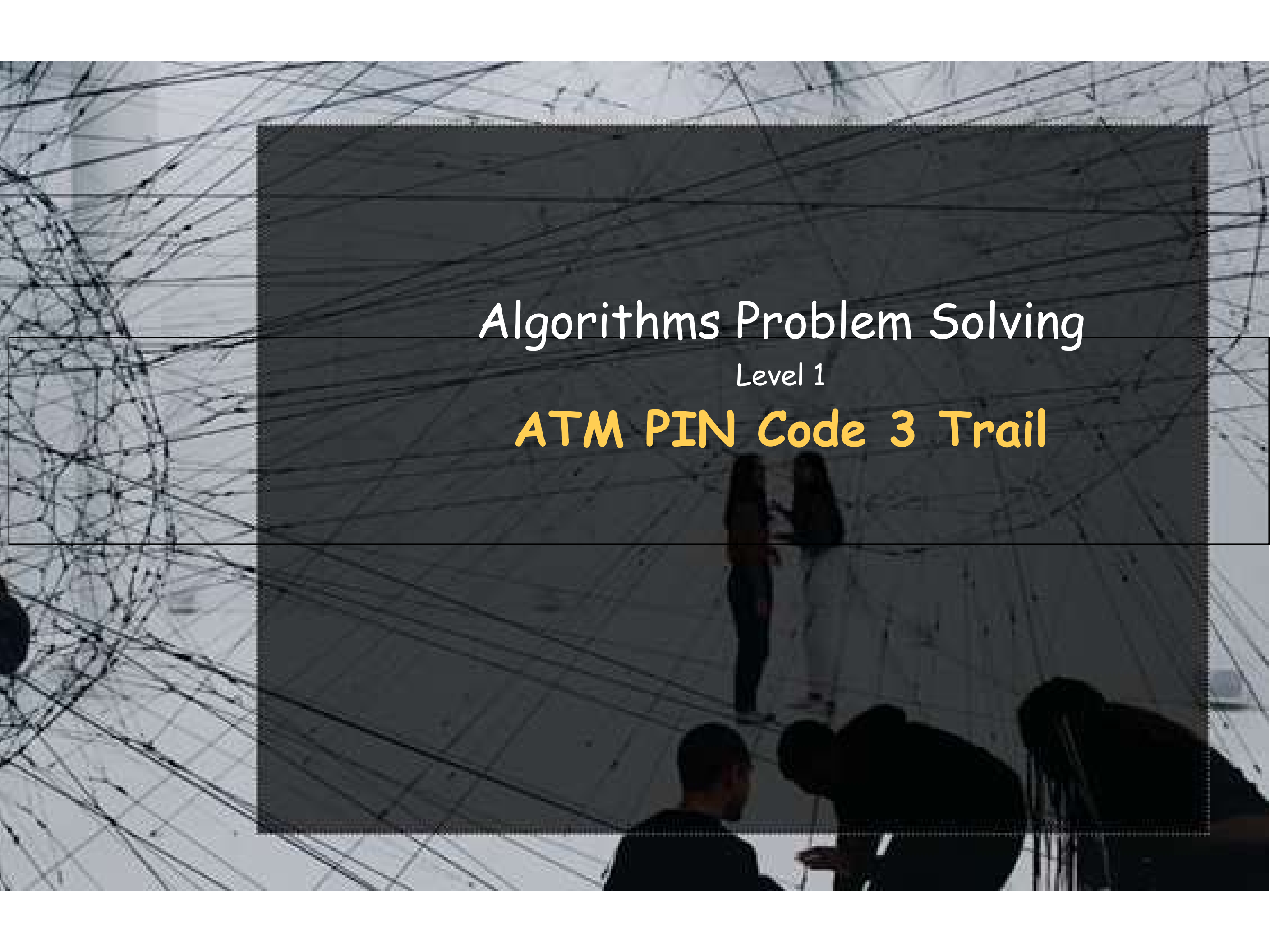


Algorithms Problem Solving

Level 1

ATM PIN Code 3 Trail



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Definition

- ◆ This program asks the user to enter an ATM PIN code and allows a maximum of three attempts.
- ◆ If the user enters the correct PIN `1234`, the program displays the account balance.
- ◆ If the user enters the PIN incorrectly three times, the program displays that the card is blocked.
- ◆ The program uses a constant, a `while` loop, a counter, a string, and conditional statements.

Explanation

- ◆ ``ReadPinCode()`` reads the PIN from the user as a ``string``.
- ◆ ``CorrectPin`` is a constant string containing the correct PIN:
``const string CorrectPin = "1234";``
- ◆ ``Trials`` keeps track of how many incorrect attempts have been made.
- ◆ The ``while (Trials < 3)`` loop allows a maximum of three attempts.
- ◆ If the entered PIN matches ``CorrectPin``, the balance is displayed and ``return`` immediately exits the function.
- ◆ If the PIN is incorrect, ``Trials`` is increased by 1 using:
``Trials++;``
- ◆ The program displays the current number of failed attempts.
- ◆ After three incorrect attempts, the loop ends and the program displays:
``Your card is blocked.``

Steps in main()

1. Set the correct PIN to `"1234"`.
2. Initialize `Trials`` to 0.
3. Read the user's PIN.
4. Compare the entered PIN with the correct PIN.
5. If correct, display the balance and exit.
6. If incorrect, increase `Trials`` by 1.
7. Repeat while `Trials < 3``.
8. If all three attempts are incorrect, display the blocked-card message.

💡 Example

◆ If the user enters:

1111

2222

1234

◆ The first two attempts are incorrect.

◆ The third attempt is correct.

➡ Result

Invalid PIN Code (1/3)

Invalid PIN Code (2/3)

Your account balance is \$7500

Note

- ◆ `Trials` counts only incorrect attempts.
- ◆ The correct PIN can be entered on the first, second, or third attempt.
- ◆ `return` stops the function immediately when the correct PIN is entered.
- ◆ `const` means `CorrectPin` cannot be changed after it is initialized.

⊘ Warning

◆ The program does not prevent the user from entering non-numeric characters because the PIN is stored as a string. This is acceptable for a PIN code, but there is no additional format validation such as requiring exactly four digits.



Comparison

▨ Correct PIN

- ✓ The entered PIN equals ` "1234" `.
- ✓ The balance is displayed.
- ✓ The function ends immediately.
- ✓ No further attempts are allowed.

▨ Incorrect PIN

- ✓ ` Trials ` increases by 1.
- ✓ An invalid PIN message is displayed.
- ✓ The user can try again if fewer than 3 attempts have been made.
- ✓ After 3 incorrect attempts, the card is blocked.

✦ Summary

- ✓ A `while` loop can limit the number of PIN attempts.
- ✓ A counter can track incorrect attempts.
- ✓ `const` can be used for a value that should not change.
- ✓ `return` can immediately exit a function when the correct PIN is entered.
- ✓ After three incorrect attempts, the program blocks the card.

Thank you 😊